

Diagonalizable groups

SGA 3, Exposé VIII

English draft

A. Grothendieck

About this draft

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Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a % **typo:** comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked % **typo?:**. Scholarly proofreading remains outstanding.

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Contents

About this draft	iii
Exposé VIII. Diagonalizable groups	1
1. Biduality	1
2. Schematic properties of diagonalizable groups	6
3. Exactness properties of the functor D_S	7
4. Torsors under a diagonalizable group	8
5. Quotient of an affine scheme by a diagonalizable group acting freely	11
6. Essentially free morphisms, and representability of certain functors of the form $\prod_{Y/S} Z/Y$	14
7. Appendix: On monomorphisms of group preschemes	18
Bibliography	24

Exposé VIII. Diagonalizable groups

by A. Grothendieck

1. Biduality

Let $\mathcal{C}$ be a category, which we identify as usual with a full subcategory of $\widehat{\mathcal{C}} = \text{Hom}(\mathcal{C}, (\text{Ens}))$ (cf. Exp. I). Let I be a commutative group functor, i.e. an object of $\widehat{\mathcal{C}}$ endowed with a commutative group structure (cf. I, 2.1).⁽¹⁾ For every $X \in \text{Ob}(\widehat{\mathcal{C}})$, the object $\underline{\text{Hom}}(X, I)$ is endowed with a commutative group structure, induced by that of I . For every group G in $\widehat{\mathcal{C}}$, let

$$D(G) = \underline{\text{Hom}}_{\text{gr.}}(G, I)$$

be the subobject of $\underline{\text{Hom}}(G, I)$ defined, for every $S \in \text{Ob}(\mathcal{C})$, by:

$$(x) \quad D(G)(S) = \text{Hom}_{S\text{-gr.}}(G_S, I_S),$$

where $G_S = G \times S$ and $I_S = I \times S$ are regarded as S -groups, i.e. groups in $\widehat{\mathcal{C}}/S$. Then $D(G)$ is a sub- $\widehat{\mathcal{C}}$ -group of $\underline{\text{Hom}}(G, I)$. In this way one obtains a contravariant functor D from the category of $\widehat{\mathcal{C}}$ -groups to the category of commutative $\widehat{\mathcal{C}}$ -groups.

The right-hand side of (x) may also be interpreted as the subset of $\text{Hom}(G \times S, I)$ consisting of the morphisms $G \times S \rightarrow I$ that are “multiplicative with respect to the first argument G ”. Moreover, the preceding formulas hold more generally when S is any object of $\widehat{\mathcal{C}}$, not necessarily coming from $\mathcal{C}$.

If one now takes for S a group in $\widehat{\mathcal{C}}$, which we shall denote by G' , then in the left-hand side $\text{Hom}(G', D(G))$ of (x) one may distinguish the subset $\text{Hom}_{\text{gr.}}(G', D(G))$ consisting of morphisms that respect the group structures of G' and $D(G)$. It then corresponds to the subset of $\text{Hom}(G \times G', I)$ consisting of morphisms that are multiplicative with respect to the first and with respect to the second argument, — which one may call bilinear morphisms from $G \times G'$ to I , or pairings of G and G' with values in I . One thus finds

$$(xx) \quad \text{Hom}_{\text{gr.}}(G', D(G)) \xrightarrow{\sim} \text{Hom}_{\text{bil.}}(G \times G', I),$$

which is an isomorphism functorial in the pair (G, G') . As the right-hand side is symmetric in G and G' , one deduces a functorial bijection:

$$(xxx) \quad \text{Hom}_{\text{gr.}}(G', D(G)) \xrightarrow{\sim} \text{Hom}_{\text{gr.}}(G, D(G')).$$

In other words, “it amounts to the same thing” to give a group homomorphism $G' \rightarrow D(G)$, or a group homomorphism $G \rightarrow D(G')$, both indeed amounting to giving a pairing $G \times G' \rightarrow I$.

Applying this to the case where $G' = D(G)$ and to the identity homomorphism $G' \rightarrow D(G)$, one then finds a *canonical homomorphism*

$$(xxxx) \quad G \rightarrow D(D(G)).$$

⁽⁰⁾Version 1.1 of November 8, 2009: additions in 1.2, 1.4, 1.7, 3.1, 3.4, 4.5.1, 6.4, 6.8 — 1.5.1 and section 7 to be reviewed.

⁽¹⁾**N.D.E.** The original has been expanded in what follows.

Definition 1.0. ⁽²⁾ One shall say that G is *reflexive* (with respect to I) if the preceding homomorphism is an isomorphism. One will note that this implies that G is commutative.

One thus sees that:

Proposition 1.0.1. *The functor D induces an anti-equivalence of the category of reflexive $\widehat{\mathcal{C}}$ -groups with itself.*

In particular, if G, H are two reflexive groups, D induces an isomorphism:

$$\mathrm{Hom}_{\mathrm{gr.}}(G, H) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{gr.}}(D(H), D(G))$$

(it even suffices for H to be reflexive, as one sees from formula (xxx)).

Definition 1.0.2. As usual, we shall then say that a $\mathcal{C}$ -group G is reflexive if it is reflexive as a $\widehat{\mathcal{C}}$ -group (without concerning ourselves with whether $D(G)$ is representable or not).

One thus obtains, by D , an anti-equivalence of the category of reflexive $\mathcal{C}$ -groups G such that $D(G)$ is representable, with itself.

Remark 1.0.3. Let us point out, to conclude these generalities, that the formation of the duals $D(G)$ commutes with extension of the base, which therefore transforms reflexive groups into reflexive groups.

We shall henceforth be interested in the case where $\mathcal{C} = (\mathrm{Sch})/S$, the category of preschemes over S , and $I = \mathbf{G}_{\mathrm{m},S}$, the “multiplicative group over S ”, cf. exposé I. For every ordinary group M , we consider the S -group M_S . One sees immediately that for every group prescheme J over S , one has a canonical isomorphism (functorial in M and J , and compatible with extension of the base):

$$\mathrm{Hom}_{S\text{-gr.}}(M_S, J) = \mathrm{Hom}_{\mathrm{gr.}}(M, J(S)).$$

Applying this to $J = I = \mathbf{G}_{\mathrm{m},S}$ and to a variable S' over S , one finds a functorial isomorphism:

$$(1.0.4) \quad D(M_S)(S') \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{gr.}}(M, \mathbf{G}_{\mathrm{m}}(S')).$$

One thus recovers the functor already considered in I, 4.4, also denoted by $D_S(M)$, which is representable for commutative M , since

$$D_S(M) = D(M_S) = \mathrm{Spec} \mathcal{O}_S(M),$$

where $\mathcal{O}_S(M)$ denotes the group algebra of M with coefficients in $\mathcal{O}_S$. (Let us also note, in the general case, that $D(M)$ does not change if one replaces M by its abelianization, so that nothing is lost by supposing M commutative.)

Definition 1.1. A group prescheme G over S is said to be *diagonalizable* if it is isomorphic to a scheme of the form $D_S(M) = D(M_S) = \underline{\mathrm{Hom}}_{S\text{-gr.}}(M_S, \mathbf{G}_{\mathrm{m}})$ for a suitable commutative group M .

One says that G is *locally diagonalizable* if every point of S admits an open neighborhood U such that $G|U$ is diagonalizable.

Theorem 1.2. *Let Γ be a constant commutative group scheme over S , i.e. isomorphic to a group scheme of the form M_S , where M is an ordinary commutative group. Then Γ is reflexive, i.e. the canonical homomorphism*

$$\Gamma \longrightarrow D(D(\Gamma))$$

is an isomorphism. ⁽³⁾ *The diagonalizable group $D(M_S)$ is therefore also reflexive.*

⁽²⁾**N.D.E.** The numbering 1.0, 1.0.1, ... has been added in order to bring out the definitions and statements found there.

⁽³⁾**N.D.E.** The following sentence has been added.

In view of the definitions, this follows from the following statement (which one shall apply to a prescheme S' over S):

Corollary 1.3. *Let $G = D(M_S)$. Then every homomorphism of S -groups*

$$u : G \longrightarrow \mathbf{G}_{m,S}$$

is defined by a uniquely determined section of M_S over S , i.e. by a uniquely determined locally constant map from S to M .

PROOF. Since by definition one has

$$\mathbf{G}_{m,S} = \mathrm{GL}(1)_S = \underline{\mathrm{Aut}}_{\mathcal{O}_S\text{-mod.}}(\mathcal{O}_S),$$

one sees that giving a group homomorphism $G \longrightarrow \mathbf{G}_{m,S}$ is equivalent to giving on $\mathcal{O}_S$ a G - $\mathcal{O}_S$ -module structure compatible with the natural $\mathcal{O}_S$ -module structure on $\mathcal{O}_S$ (cf. I, 4.7). By I, 4.7.3, this also amounts to giving a grading of type M on $\mathcal{O}_S$, i.e. a decomposition of $\mathcal{O}_S$ as a direct sum of modules $\mathcal{L}_m$ ($m \in M$). Now it is well known that a direct summand of a locally free module of finite type is locally free of finite type, so each $\mathcal{L}_m$ is, in a neighborhood of each point of S , either zero or free of rank 1, and in the latter case identical to $\mathcal{O}_S$ in this neighborhood. Let S_m be the open subset of S consisting of the points where this second alternative occurs. Expressing the fact that $\mathcal{O}_S$ is the direct sum of the $\mathcal{L}_m$, one sees that the union of the S_m is S and that the S_m are pairwise disjoint. Thus giving a group homomorphism $G \longrightarrow \mathbf{G}_{m,S}$ is equivalent to giving a decomposition of S as a union of pairwise disjoint open subsets S_m ($m \in M$), i.e. to giving a locally constant map from S to M . This establishes 1.3 and hence 1.2. $\square$

Corollary 1.4. *A diagonalizable group is reflexive; the same is therefore true of a locally diagonalizable group. If M, N are two ordinary commutative groups, the natural homomorphism*

$$\mathrm{Hom}_{S\text{-gr.}}(M_S, N_S) \longrightarrow \mathrm{Hom}_{S\text{-gr.}}(D(N_S), D(M_S))$$

is bijective.

(4) As the preceding isomorphism is compatible with extension of the base, one deduces an isomorphism of S -group functors:

$$(1) \quad \underline{\mathrm{Hom}}_{S\text{-gr.}}(M_S, N_S) \xrightarrow{\sim} \underline{\mathrm{Hom}}_{S\text{-gr.}}(D(N_S), D(M_S)).$$

For every S -prescheme T , one has

$$\underline{\mathrm{Hom}}_{S\text{-gr.}}(M_S, N_S)(T) = \mathrm{Hom}_{\mathrm{gr.}}(M, \Gamma(N_T/T)),$$

and, by I 1.8, $\Gamma(N_T/T)$ is the abelian group of locally constant maps $T \longrightarrow N$. On the other hand, let $\mathrm{Hom}_{\mathrm{gr.}}(M, N)_S$ be the constant S -group associated with the ordinary abelian group $\mathrm{Hom}_{\mathrm{gr.}}(M, N)$. One has an evident homomorphism of commutative S -group functors:

$$(2) \quad \mathrm{Hom}_{\mathrm{gr.}}(M, N)_S \xrightarrow{\theta} \underline{\mathrm{Hom}}_{S\text{-gr.}}(M_S, N_S),$$

which is always a monomorphism. Moreover, it is an *isomorphism* if M is of finite type.⁽⁵⁾

⁽⁴⁾**N.D.E.** What follows has been expanded.

⁽⁵⁾**N.D.E.** Indeed, denoting by $F(M)$ (resp. $G(M)$) the left-hand (resp. right-hand) side of (2), if $M = M_1 \oplus M_2$, one has a canonical isomorphism $F(M) = F(M_1) \oplus F(M_2)$, and similarly for G . This reduces us to checking that θ is an isomorphism when $M = \mathbf{Z}/r\mathbf{Z}$, for an integer $r \geq 0$. In this case, $F(M) = ({}_rN)_S$, where ${}_rN$ is the kernel of $r \cdot \mathrm{id}_N$, and, for every $T \longrightarrow S$, the homomorphism

$$F(M)(T) = \Gamma({}_rN_T/T) \longrightarrow G(M)(T) = {}_r\Gamma(N_T/T)$$

is bijective, whence the desired result.

Part (a) of the following corollary follows from the above; part (b) follows from it by the descent results “recalled” in 1.7.⁽⁶⁾

Corollary 1.5. *a) Let M, N be two ordinary commutative groups, M of finite type; then one has an isomorphism*

$$\mathrm{Hom}_{\mathrm{gr.}}(M, N)_S \xrightarrow{\sim} \underline{\mathrm{Hom}}_{S\text{-gr.}}(D(N_S), D(M_S));$$

consequently $\underline{\mathrm{Hom}}_{S\text{-gr.}}(D(N_S), D(M_S))$ is representable.

b) More generally, if G, H are locally diagonalizable, H being of finite type, then $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$ is representable.

⁽⁷⁾ One concludes from 1.5:

Corollary 1.6. *Under the conditions of 1.5, if S is connected, one has*

$$\mathrm{Hom}_{S\text{-gr.}}(D_S(N), D_S(M)) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{gr.}}(M, N)$$

and

$$\mathrm{Isom}_{S\text{-gr.}}(D_S(N), D_S(M)) \xrightarrow{\sim} \mathrm{Isom}_{\mathrm{gr.}}(M, N).$$

1.7. — Descent of representability. — ⁽⁸⁾ In this paragraph one “recalls” some descent results which will frequently be used in what follows.

Scholium 1.7.1. *Let S be a prescheme and X, Y, T be S -preschemes. If (T_i) is an open covering of T , and if one puts $T_{ij} = T_i \cap T_j = T_i \times_T T_j$, then, since giving a morphism of T -preschemes $X_T \rightarrow Y_T$ is local on T , one has an exact sequence of sets:*

$$(1) \quad \mathrm{Hom}_T(X_T, Y_T) \rightarrow \prod_i \mathrm{Hom}_{T_i}(X_{T_i}, Y_{T_i}) \rightrightarrows \prod_{i,j} \mathrm{Hom}_{T_{ij}}(X_{T_{ij}}, Y_{T_{ij}})$$

i.e. $\underline{\mathrm{Hom}}_S(X, Y)$ is a local S -functor, that is, a sheaf on $(\mathrm{Sch})_S$ endowed with the Zariski topology.

More generally, by IV 4.5.13, $\underline{\mathrm{Hom}}_S(X, Y)$ is a sheaf on $(\mathrm{Sch})_S$ for every topology coarser than the canonical topology, for example for the (fpqc) topology.

If G, H are S -group preschemes, one deduces that the subfunctor $\underline{\mathrm{Hom}}_{S\text{-gr.}}(X, Y)$ is a sheaf for the (fpqc) topology (and hence a fortiori a local functor).

Lemma 1.7.2. ⁽⁹⁾ *Let F be a local S -functor.*

(i) Suppose that there exists an open covering (S_i) of S such that the restriction $F_i = F \times_S S_i$ of F to each S_i is representable by an S_i -prescheme X_i . Then F is representable by an S -prescheme X .

(ii) Suppose that F is an (fpqc) sheaf and that there exists a faithfully flat and quasi-compact morphism $S' \rightarrow S$ such that the restriction $F' = F \times_S S'$ of F is representable by an S' -prescheme X' . Then X' is endowed with a descent datum (cf. IV 2.1) with respect to $S' \rightarrow S$.

If, in addition, this descent datum is effective (which is the case if X' is affine over S'), then F is representable by an S -prescheme X .

⁽⁶⁾**N.D.E.** After 1.5, the original stated: “One concludes more generally that if G, H are locally diagonalizable, H being of finite type, then $\underline{\mathrm{Hom}}_{S\text{-gr.}}(G, H)$ is representable”. This assertion has been included in the statement of 1.5, and its proof has been made explicit in 1.7.

⁽⁷⁾**N.D.E.** A statement 1.5.1 dealing with the functor $\underline{\mathrm{Isom}}_{S\text{-gr.}}(G, H)$, considered in X 5.10 and 5.11, ought to be added. . .

⁽⁸⁾**N.D.E.** This paragraph has been added to make explicit the “local on S ” character of the representability of certain sheaves over S , used many times in what follows (and implicitly in the original).

⁽⁹⁾**N.D.E.** See also remark XI 3.4.

PROOF. (i) It follows from the hypothesis that $X_i \times_S S_j$ and $X_j \times_S S_i$ both represent the restriction of F to $S_{ij} = S_i \times_S S_j$, so, by Yoneda's lemma, there exists a unique isomorphism of S_{ij} -preschemes

$$c_{ji} : X_i \times_S S_j \xrightarrow{\sim} X_j \times_S S_i;$$

one then has isomorphisms of preschemes over $S_{ijk} = S_i \times_S S_j \times_S S_k$:

$$\begin{array}{ccc} X_i \times_S S_j \times_S S_k & \xrightarrow{c_{ji} \times \text{id}_{S_k}} & X_j \times_S S_i \times_S S_k = X_j \times_S S_k \times_S S_i \\ \parallel & & \downarrow c_{kj} \times \text{id}_{S_i} \\ X_i \times_S S_k \times_S S_j & \xrightarrow{c_{ki} \times \text{id}_{S_j}} & X_k \times_S S_i \times_S S_j = X_k \times_S S_j \times_S S_i \end{array}$$

and as all these objects represent the restriction of F to S_{ijk} , this diagram is commutative, i.e. the c_{ji} satisfy the usual cocycle relation $c_{kj} \circ c_{ji} = c_{ki}$.

It follows that the X_i glue into an S -prescheme X such that $X \times_S S_i = X_i$ for every i . Thus, for every Y over S_i , one has

$$(*) \quad F(Y) = F_i(Y) = \text{Hom}_{S_i}(Y, X \times_S S_i) = \text{Hom}_S(Y, X) = h_X(Y).$$

Then, for arbitrary $Y \rightarrow S$, the $Y_i = Y \times_S S_i$ form an open covering of Y ; put $Y_{ij} = Y_i \times_Y Y_j = Y \times_S S_{ij}$. Since F (resp. h_X) is a local functor by hypothesis (resp. since the Zariski topology is coarser than the canonical topology), $F(Y)$ and $h_X(Y)$ are both identified, in view of (*), with the kernel of the double arrow:

$$\begin{array}{ccc} \prod_i F(Y_i) & \rightrightarrows & \prod_{i,j} F(Y_{ij}) \\ \parallel & & \parallel \\ \prod_i h_X(Y_i) & \rightrightarrows & \prod_{i,j} h_X(Y_{ij}). \end{array}$$

This proves (i).

(ii) It follows from the hypothesis that $F'_1 = F' \times_{S'} S''_1$ (where $S''_1 = S'' = S' \times_S S'$ regarded as an S' -prescheme via the first projection) is represented by $X''_1 = X' \times_{S'} S''_1$; similarly, $F''_2 = F' \times_{S'} S''_2$ is represented by $X''_2 = X' \times_{S'} S''_2$. Now $F'_1 = F' \times_S S'' = F''_2$, so there exists a (unique) S'' -isomorphism $c : X''_1 \xrightarrow{\sim} X''_2$; then, denoting by q_i (resp. p_{ji}) the projection of $S''' = S' \times_S S' \times_S S'$ onto the i -th factor (resp. onto the factors i and j), by $X'''_i = X' \times_{S'} S'''_i$ (where $S'''_i = S'''$ regarded as an S' -prescheme via q_i), and by $p^*_{ji}(c) : X'''_i \xrightarrow{\sim} X'''_j$ the isomorphism of S''' -preschemes deduced from c by base change, one obtains a diagram of isomorphisms of S''' -preschemes:

$$\begin{array}{ccc} X'''_1 & \xrightarrow{p^*_{21}(c)} & X'''_2 \\ & \searrow p^*_{31}(c) & \downarrow p^*_{32}(c) \\ & & X'''_3 \end{array}$$

and as all these objects represent the restriction of F to S''' , this diagram is commutative, i.e. one has the usual cocycle relation $p^*_{32}(c) \circ p^*_{21}(c) = p^*_{31}(c)$, i.e. c is a descent datum on X' with respect to $S' \rightarrow S$ (cf. IV 2.1). Suppose, in addition, that this descent datum is effective, i.e. that there exists an S -prescheme X such that $X' \simeq X \times_S S'$ (by SGA 1, VIII 2.1, this is the case if X' is affine over $S'^{(10)}$). Then, for every $Y \rightarrow S'$, one has

$$(**) \quad F(Y) = F'(Y) = \text{Hom}_{S'}(Y, X \times_S S') = \text{Hom}_S(Y, X) = h_X(Y).$$

⁽¹⁰⁾ **N.D.E.** For another criterion of effectiveness, see below X 5.4–5.6.

Then, for arbitrary $Y \rightarrow S$, put $Y' = Y \times_S S'$ and $Y'' = Y' \times_Y Y' \simeq Y \times_S S''$. Then $Y' \rightarrow Y$ is, like $S' \rightarrow S$, faithfully flat and quasi-compact, hence an $\mathcal{M}$ -effective epimorphism (where $\mathcal{M}$ = the family of faithfully flat quasi-compact morphisms), i.e. the equivalence relation

$$Y' \times_Y Y' \rightrightarrows Y'$$

is $\mathcal{M}$ -effective and has quotient Y . Since F (resp. h_X) is an (fpqc) sheaf by hypothesis (resp. since the (fpqc) topology is coarser than the canonical topology), $F(Y)$ and $h_X(Y)$ are both identified, in view of (**), with the kernel of the double arrow:

$$\begin{array}{ccc} F(Y') & \rightrightarrows & F(Y' \times_Y Y') \\ \parallel & & \parallel \\ h_X(Y') & \rightrightarrows & h_X(Y' \times_Y Y'). \end{array}$$

This proves (ii). $\square$

Corollary 1.7.3. *Let F be an (fpqc) sheaf on $(\text{Sch})_S$. Suppose that there exists an open covering (S_i) of S and, for each i , a faithfully flat and quasi-compact morphism $S'_i \rightarrow S_i$ such that the restriction $F'_i = F \times_S S'_i$ is representable by an S'_i -prescheme X'_i affine over S'_i . Then F is representable by an S -prescheme X affine over S (such that $X \times_S S'_i = X'_i$ for every i).*

If, in addition, each $X'_i \rightarrow S'_i$ is a closed immersion (resp. a finite étale morphism), the same is true of $X \rightarrow S$.

The first assertion follows from 1.7.2. For the second, it suffices to check that each morphism $X \times_S S_i \rightarrow S_i$ is a closed immersion (resp. finite and étale), which follows from EGA IV₂, 2.7.1 (resp. and IV₄, 17.7.3).

Remark 1.7.4. Assertion 1.5 (b) follows, as announced, from 1.7.1 and 1.7.2 (i).

2. Schematic properties of diagonalizable groups

They are summarized in the

Proposition 2.1. *Let S be a nonempty prescheme, M an ordinary commutative group, $G = D(M_S)$ the diagonalizable S -group defined by M . One has the following:*

- a) G is faithfully flat over S , and affine over S (a fortiori quasi-compact over S).
- b) M of finite type $\iff G$ of finite type over $S \iff G$ of finite presentation over S .
- c) M finite $\iff G$ finite over $S \iff G$ of finite type over S and annihilated by an integer $n > 0$. Then $\deg(G/S) = \text{Card}(M)$.
- c') M a torsion group $\iff G$ integral over S .
- d) $M = 0 \iff G =$ the unit S -group.
- e) M of finite type, and the order of its torsion subgroup prime to the residue characteristics of $S \iff G$ smooth over S .

The verification of a) to d) is trivial and left to the reader. Let us prove e). If G is smooth over S , it is locally of finite presentation over S , hence of finite presentation over S since it is affine over S , so M is of finite type. One may therefore already suppose M of finite type, hence G of finite presentation over S . Then⁽¹¹⁾ G is smooth over S if and only if its geometric fibers are so, which reduces us to the case where S is the spectrum of an algebraically closed field k . Writing $M = T \times L$, with T the torsion subgroup and L free, $L \simeq \mathbf{Z}^r$, one has $D(M) = D(T) \times D(L)$, where $D(L) \simeq \mathbf{G}_m^r$ is smooth over k . Thus $G = D(M)$ is smooth over k if and only if $D(T)$ is so, which means, since $D(T)$ is finite over k of degree equal to the order n of T , that $D(T)(k)$ has n elements. Now T is isomorphic to a sum of groups

⁽¹¹⁾**N.D.E.** (since G is flat over S , by a))

$\mathbf{Z}/n_i\mathbf{Z}$, n being the product of the n_i , so $D(T)$ is a product of the $D(\mathbf{Z}/n_i\mathbf{Z}) = \mu_{n_i}$ (group schemes of n_i -th roots of unity), hence

$$\text{card}(D(T)(k)) = \prod_i \text{card } \mu_i(k)$$

where $\text{card } \mu_i(k) = (\text{number of } n_i\text{-th roots of unity in } k) \leq n_i$, equality being attained if and only if n_i is prime to the characteristic p of k . Thus one has $\text{card}(D(T)(k)) = n$ (where $n = \prod_i n_i$) if and only if all the n_i are prime to p , i.e. if and only if n is prime to p . Q.E.D.

3. Exactness properties of the functor D_S

Theorem 3.1. *Let S be a prescheme, and*

$$0 \longrightarrow M' \xrightarrow{u} M \xrightarrow{v} M'' \longrightarrow 0$$

an exact sequence of ordinary commutative groups. Consider the sequence of transposed homomorphisms:

$$0 \longrightarrow D_S(M'') \xrightarrow{v^t} D_S(M) \xrightarrow{u^t} D_S(M') \longrightarrow 0.$$

(i) v^t induces an isomorphism of $D_S(M'')$ with the kernel of u^t , and u^t is faithfully flat and quasi-compact.

(ii)⁽¹²⁾ $D_S(M')$ represents the (fpqc) quotient sheaf $D_S(M)/D_S(M'')$.

Denote by $\mathcal{M}$ the family of faithfully flat quasi-compact morphisms. First, (ii) follows from (i) (cf. IV, 4.6.5.1). Indeed, the equivalence relation in $D_S(M)$ defined by u^t is the same as that defined by the subgroup $\text{Ker}(u^t) = D_S(M'')$; as $u^t \in \mathcal{M}$, this equivalence relation is $\mathcal{M}$ -effective (cf. IV, 3.3.2.1), and therefore $D_S(M')$ represents the quotient sheaf for the (fpqc) topology (cf. IV, 4.6.5).

The first assertion of (i) is a trivial consequence of the definition of the functors $D_S(-)$; more generally, for every exact sequence

$$M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

(without a zero on the left), one shall have an exact transposed sequence:

$$0 \longrightarrow D_S(M'') \longrightarrow D_S(M) \longrightarrow D_S(M').$$

(This holds more generally in the context of the beginning of no. 1.) On the other hand, as $D_S(M)$ and $D_S(M')$ are affine over S , u^t is necessarily an affine morphism, a fortiori quasi-compact (whatever the homomorphism $u : M' \longrightarrow M$). Thus the second assertion of (i) will follow from part a) in the

Corollary 3.2. *Let S be a nonempty prescheme, $u : M' \longrightarrow M$ a homomorphism of ordinary commutative groups, $u^t : G \longrightarrow G'$ the transposed homomorphism. Then:*

- a) *For u to be a monomorphism, it is necessary and sufficient that u^t be faithfully flat.*
- b) *For u to be an epimorphism, it is necessary and sufficient that u^t be a monomorphism (and then u^t is even a closed immersion).*

To prove a), one notes that if u is a monomorphism, then $\mathcal{O}_S(M)$ is a module over $\mathcal{O}_S(M')$ admitting a nonempty basis (namely the system of sections defined by any system of representatives of M modulo M'), a fortiori it is faithfully flat. Conversely, if this is so, then $u^t : \mathcal{O}_S(M') \longrightarrow \mathcal{O}_S(M)$ is injective, which (for $S \neq \emptyset$) implies that $u : M' \longrightarrow M$ is injective.

To prove b), one notes that if u is an epimorphism, then $\mathcal{O}_S(M') \longrightarrow \mathcal{O}_S(M)$ is surjective, so u^t is a closed immersion and a fortiori a monomorphism. Conversely, if this is so, then $\text{Ker } u^t = \text{the unit group}$; now putting $M'' = \text{Coker } u$, one has seen that $\text{Ker } u^t \simeq D_S(M'')$, so by 2.1 d) one has $M'' = 0$, hence u is an epimorphism.

⁽¹²⁾**N.D.E.** The following has been added, and the proof has been expanded accordingly.

One concludes from 3.1 in the usual way:

Corollary 3.3. *Let $M' \xrightarrow{u} M \xrightarrow{v} M''$ be an exact sequence of ordinary commutative groups; consider the transposed sequence*

$$G'' \xrightarrow{v^t} G \xrightarrow{u^t} G'.$$

Then v^t induces a faithfully flat and quasi-compact morphism from G'' to $\text{Ker } u^t$, and the latter is a diagonalizable group isomorphic to $D_S(v(M)) = D_S(\text{Coker } u)$.

Corollary 3.4. *Let S be a prescheme, $u : G \rightarrow H$ a homomorphism of locally diagonalizable S -group preschemes, with H of finite type over S . Put $G' = \text{Ker } u$. Then:*

- a) G' is locally diagonalizable; it is of finite type over S if G is so.*
- b) The quotient G/G' “exists”; more precisely, the equivalence relation defined by G' in G is $\mathcal{M}$ -effective (where $\mathcal{M}$ = the set of faithfully flat quasi-compact morphisms, cf. IV, 3.4). Moreover, G/G' is locally diagonalizable, of finite type over S .*
- c) The homomorphism $u : G \rightarrow H$ factors uniquely as*

$$G \xrightarrow{v} G/G' \xrightarrow{w} H,$$

where v is the canonical homomorphism (so v is faithfully flat and quasi-compact). Moreover, w is a closed immersion, and a fortiori a monomorphism.

Finally, the quotient $H' = H/\text{Im } w = \text{Coker } w = \text{Coker } u$ exists; more precisely, the equivalence relation defined by G/G' in H is $\mathcal{M}$ -effective, and H' is of finite type over S .

The first assertion of c) is a consequence of b), by definition of the quotient G/G' (cf. IV, 3.2.3).⁽¹³⁾ Let us show that the (fpqc) quotient sheaf $\widetilde{G}/\widetilde{G'}$ is representable. This is checked locally on S , as are all the other assertions; one may therefore suppose G and H diagonalizable, of the form $D_S(M)$ and $D_S(N)$.

As H is of finite type over S , N is of finite type by 2.1 b), so by 1.5, u is defined by a homomorphism $u' : N \rightarrow M$. Then, by 3.1 and 3.2, G' is isomorphic to $D_S(\text{Coker } u')$ and $\widetilde{G}/\widetilde{G'}$ is representable by $D_S(\text{Im } u')$; moreover, considering the exact sequence

$$0 \rightarrow \text{Ker } u' \rightarrow N \xrightarrow{u'} \text{Im } u' \rightarrow 0,$$

one obtains that w is a closed immersion and that the quotient $H' = H/\text{Im } w$ is $D_S(\text{Ker } u')$; this is of finite type over S since N , and hence $\text{Ker } u'$, is of finite type.

Remarks. The existence result for quotients 3.4 will be substantially generalized in no. 5.

On the other hand, one will note that in the present no. and the preceding one, the hypothesis $S \neq \emptyset$ entered only to ensure the validity of certain converses, allowing properties of the corresponding ordinary groups to be deduced from certain hypotheses on diagonalizable S -groups. The results in the “direct” sense hold without restriction on S , and the proofs given here apply in the general case.

Corollary 3.5. *Let G be a diagonalizable group prescheme over S , and let n be an integer $\neq 0$. Then the subgroup ${}_nG$ of G , kernel of the homomorphism $n \cdot \text{id}_G : G \rightarrow G$, is integral over S , and finite over S if G is of finite type over S .*

Indeed, if $G = D_S(M)$, then ${}_nG = D_S(M/nM)$ by 3.1, and one concludes by 2.1 b), c), c').

4. Torsors under a diagonalizable group

Let S be a prescheme, and $G = D_S(M)$ a diagonalizable group over S . We propose to determine the G -torsors (or principal homogeneous G -bundles) over S , in the sense of the “faithfully flat quasi-compact topology”, cf. Exp. IV, 5.1. Recall that a prescheme P

⁽¹³⁾**N.D.E.** The original has been expanded in what follows.

over S , with group of operators G , is said to be a *torsor* or *principal homogeneous* if every point of S admits an open neighborhood U and a faithfully flat and quasi-compact morphism $S' \rightarrow U$ such that $P' = P \times_S S'$ is a bundle with operators isomorphic to $G' = G \times_S S'$ (acting on itself by right translations). Since G is affine over S , it follows by SGA 1, VIII 5.6 that P is necessarily affine over S . Let us also note that since G is itself faithfully flat and quasi-compact over S , P is principal homogeneous under G if and only if it is “formally principal homogeneous” and is, in addition, faithfully flat and quasi-compact over S (cf. IV, 5.1.6).

Recall, on the other hand (Exp. I, 4.7.3), that giving an S -prescheme P affine over S with group of operators $G = D_S(M)$ amounts to giving a quasi-coherent commutative algebra graded of type M over S , i.e. a quasi-coherent algebra $\mathcal{A}$ over S endowed with a decomposition as a direct sum (as a module):

$$\mathcal{A} = \coprod_{m \in M} \mathcal{A}_m,$$

with

$$\mathcal{A}_m \cdot \mathcal{A}_{m'} \subseteq \mathcal{A}_{m+m'} \quad \text{for } m, m' \in M.$$

This being so, the answer to the problem posed above is given by the

Proposition 4.1. *For the prescheme P with group of operators $G = D_S(M)$, defined by the algebra $\mathcal{A}$ graded of type M , to be a principal homogeneous bundle under G , it is necessary and sufficient that $\mathcal{A}$ satisfy the following conditions:*

- a) *For every $m \in M$, $\mathcal{A}_m$ is an invertible module over S .*
- b) *For $m, m' \in M$, the homomorphism*

$$\mathcal{A}_m \otimes_{\mathcal{O}_S} \mathcal{A}_{m'} \rightarrow \mathcal{A}_{m+m'}$$

induced by multiplication in $\mathcal{A}$ is an isomorphism.

The necessity of the conditions is immediate by descent, since they are satisfied in the case where P is the trivial principal homogeneous bundle, i.e. $\mathcal{A} = \mathcal{O}_S(M)$. For sufficiency, one notes that a) already implies that P is faithfully flat over S ; it is in any case quasi-compact over S (being affine over S), so it remains to check that it is formally principal homogeneous under G , i.e. that the well-known homomorphism

$$P \times_S G \rightarrow P \times_S P$$

is an isomorphism. Now on the affine algebras this homomorphism is made explicit as the homomorphism

$$\mathcal{A} \otimes \mathcal{A} \rightarrow \mathcal{A}(M) = \mathcal{A} \otimes \mathcal{O}_S(M)$$

which in bidegree (m, n) (where $m, n \in M$) is given by

$$x_m \otimes y_n \mapsto x_m y_n \otimes e_n.$$

From the point of view of degrees, this homomorphism is compatible with the homomorphism $M \times M \rightarrow M \times M$ given by

$$(m, n) \mapsto (m + n, n),$$

which is an isomorphism. This shows that b) expresses precisely (independently of a)) that P is formally principal homogeneous, and establishes 4.1.

Let us also note that one obtains, by faithfully flat descent:

Corollary 4.2. *The conditions of 4.1 imply that the homomorphism*

$$\mathcal{O}_S \rightarrow \mathcal{A}_0$$

is an isomorphism.

If, for example, $M = \mathbf{Z}$, then under the conditions of 4.1 one sees that $\mathcal{A}$ is essentially known when one knows $\mathcal{A}_1 = \mathcal{L}$, namely

$$\mathcal{A} \simeq \prod_{n \in \mathbf{Z}} \mathcal{L}^{\otimes n}$$

(isomorphism of graded algebras). One thus recovers the well-known result:

Corollary 4.3. *There is an equivalence between the category of principal homogeneous bundles P over S with group $\mathbf{G}_{m,S}$ and the category of invertible modules $\mathcal{L}$ over S (taking as morphisms to define both categories the isomorphisms for the structures involved). One obtains two functors quasi-inverse to one another by associating with every P the degree 1 component of its $\mathbf{Z}$ -graded affine algebra, and by associating with every $\mathcal{L}$ the spectrum of the $\mathbf{Z}$ -graded algebra $\prod_{n \in \mathbf{Z}} \mathcal{L}^{\otimes n}$.*

In particular:

Corollary 4.4. *The group of classes of principal homogeneous bundles over S with group $\mathbf{G}_{m,S}$ is isomorphic to the group $\text{Pic}(S)$ of classes of invertible modules over S , i.e. to $H^1(S, \mathcal{O}_S^\times)$.*

In view of the fact that $\mathbf{G}_{m,S}$ is the scheme of automorphisms of the module $\mathcal{O}_S$, one sees that 4.4 is equivalent to the following statement, which is one of the variants of Hilbert's "Theorem 90".

Corollary 4.5. *Every principal homogeneous bundle over S with group $\mathbf{G}_m$ is locally trivial (in the sense of the Zariski topology).*

Remark 4.5.1. One will note that the preceding statement is no longer true in general for a group such as μ_n , or for a "twisted form" of $\mathbf{G}_m$; for example, the unique twisted form of $\mathbf{G}_m$ over the field $\mathbf{R}$ of real numbers gives a 1-cohomology group equal to $\mathbf{Z}/2\mathbf{Z}$.

⁽¹⁴⁾ Indeed, let S^1 be the kernel of the norm morphism $N : \prod_{\mathbf{C}/\mathbf{R}} \mathbf{G}_{m,\mathbf{C}} \rightarrow \mathbf{G}_{m,\mathbf{R}}$; it is a $\mathbf{C}/\mathbf{R}$ -twisted form of $\mathbf{G}_{m,\mathbf{R}}$. The equation $N(z) = -1$ in $\prod_{\mathbf{C}/\mathbf{R}} (\mathbf{G}_{m,\mathbf{C}})$ defines an S^1 -torsor X over $\text{Spec}(\mathbf{R})$, locally trivial for the étale topology, but nontrivial since $X(\mathbf{R}) = \emptyset$. Let us show that $H_{\text{ét}}^1(\mathbf{R}, S^1) \simeq \mathbf{Z}/2\mathbf{Z}$. One has an exact sequence of smooth commutative algebraic $\mathbf{R}$ -groups:

$$1 \rightarrow S^1 \rightarrow \prod_{\mathbf{C}/\mathbf{R}} \mathbf{G}_{m,\mathbf{C}} \rightarrow \mathbf{G}_{m,\mathbf{R}} \rightarrow 1$$

which gives rise to a long exact sequence of étale cohomology (or Galois cohomology):

$$0 \rightarrow S^1(\mathbf{R}) \rightarrow \mathbf{C}^\times \xrightarrow{N} \mathbf{R}^\times \rightarrow H_{\text{ét}}^1(\mathbf{R}, S^1) \rightarrow H_{\text{ét}}^1(\mathbf{R}, \prod_{\mathbf{C}/\mathbf{R}} \mathbf{G}_{m,\mathbf{C}}) \rightarrow \dots$$

Now (see, for example, XXIV, 8.4), $H_{\text{ét}}^1(\mathbf{R}, \prod_{\mathbf{C}/\mathbf{R}} \mathbf{G}_{m,\mathbf{C}}) \simeq H_{\text{ét}}^1(\mathbf{C}, \mathbf{G}_{m,\mathbf{C}})$, and the latter is zero by 4.5 (or, here, since $\mathbf{C}$ is algebraically closed). One therefore obtains an isomorphism $H_{\text{ét}}^1(\mathbf{R}, S^1) \simeq \mathbf{R}^\times / N(\mathbf{C}^\times) \simeq \{\pm 1\}$.

We shall need the following result in the next no. :

Proposition 4.6. *Under the conditions of 4.1, conditions a) and b) are equivalent to the following conditions:*

a') $\mathcal{O}_S \rightarrow \mathcal{A}_0$ is an isomorphism.

b') For every m in M (it suffices: in a system of generators of M), one has:

$$\mathcal{A}_m \cdot \mathcal{A}_{-m} = \mathcal{A}_0.$$

⁽¹⁴⁾**N.D.E.** The following has been added.

Necessity being evident, in view of 4.2⁽¹⁵⁾, we shall reduce to proving the

Corollary 4.7. *Let $A = \coprod_{n \in \mathbf{Z}} A_n$ be a $\mathbf{Z}$ -graded ring such that*

$$A_1 \cdot A_{-1} = A_0.$$

Then the A_n are invertible A_0 -modules, and, for $n, n' \in \mathbf{Z}$, the homomorphism

$$A_n \otimes_{A_0} A_{n'} \longrightarrow A_{n+n'}$$

induced by multiplication in A is an isomorphism.

By hypothesis, there exist $f_i \in A_1, g_i \in A_{-1}$ such that

$$(*) \quad \sum_i f_i g_i = 1.$$

As the conclusion to be established is local on $\text{Spec}(A_0)$,⁽¹⁶⁾ and as, by (*), $\text{Spec}(A_0)$ is covered by the affine open subsets $D(f_i g_i)$, one is reduced to the case where there exists an element $f \in A_1$ invertible in A . Then, for every $n \in \mathbf{Z}$, f^n is an element of A_n invertible in A , and therefore defines an isomorphism $h \mapsto f^n h$ from A_0 onto A_n . Moreover, this shows that one obtains an isomorphism $A_0[t, t^{-1}] \rightarrow A$ of graded A_0 -algebras by sending t to f , which completes the proof of 4.7.

Then, under the conditions of 4.6, 4.7 already implies that the $\mathcal{A}_m$ ($m \in M$) are invertible. To prove condition 4.1 b), one may therefore suppose that $\mathcal{A}_m$ and $\mathcal{A}_{m'}$ have bases f_m and $f_{m'}$, with inverses $f_m^{-1} \in \Gamma(\mathcal{A}_{-m})$ and $f_{m'}^{-1} \in \Gamma(\mathcal{A}_{-m'})$. Then multiplication by $f_m^{-1} f_{m'}^{-1} \in \Gamma(\mathcal{A}_{-m-m'})$ defines a homomorphism $\mathcal{A}_{m+m'} \rightarrow \mathcal{A}_0 \simeq \mathcal{O}_S$, transforming the image of $f_m \otimes f_{m'}$ into the section 1 of $\mathcal{O}_S$. In the diagram

$$\begin{array}{ccccc} & & w & & \\ & \searrow & & \nearrow & \\ \mathcal{A}_m \otimes \mathcal{A}_{m'} & \xrightarrow{u} & \mathcal{A}_{m+m'} & \xrightarrow{v} & \mathcal{A}_0 \simeq \mathcal{O}_S \end{array}$$

w and v are therefore epimorphisms of invertible sheaves, hence isomorphisms, so u is an isomorphism. Q.E.D.

5. Quotient of an affine scheme by a diagonalizable group acting freely

⁽¹⁷⁾ Denote by $\mathcal{M}$ the set of faithfully flat quasi-compact morphisms, and recall that we consider torsors in the sense of the (fpqc) topology.

Theorem 5.1. *Let S be a prescheme, M an ordinary commutative group, $G = D_S(M)$ the diagonalizable group over S that it defines, P an S -prescheme affine over S on which G acts freely on the right.*

Then the equivalence relation defined by G in P is $\mathcal{M}$ -effective (cf. IV, 3.4), i.e. the quotient $X = P/G$ exists and P is a torsor over X with group $G_X = D_X(M)$. Moreover, P/G is affine over S ; more precisely, if P is defined by the algebra $\mathcal{A}$ graded of type M , then P/G is isomorphic to $\text{Spec}(\mathcal{A}_0)$, where $\mathcal{A}_0 = \mathcal{A}^G$ is the degree 0 component of $\mathcal{A}$.

PROOF. Put $X = \text{Spec}(\mathcal{A}_0)$; then one has a natural morphism $P \rightarrow X$, deduced from $\mathcal{A}_0 \rightarrow \mathcal{A}$, which is invariant under the operations of G . In this way P becomes an X -prescheme, affine over X , with group of operators $G_X = D_X(M)$, and the hypothesis that G acts freely on P/S implies that G_X acts freely on P/X . Everything amounts to showing that P is a principal homogeneous bundle under G_X , using the fact that $\mathcal{B}_0 = \mathcal{O}_X$, where $\mathcal{B}$ is the algebra graded of type M over X that defines P/X . One may then suppose $X = S$ and S affine, so P is affine, given by a ring A graded of type M , whose homogeneous part of degree

⁽¹⁵⁾ N.D.E. 4.1 has been corrected to 4.2.

⁽¹⁶⁾ N.D.E. The following passage has been expanded.

⁽¹⁷⁾ N.D.E. The following sentence has been added.

m will be denoted by A_m , so that $S = \text{Spec}(A_0)$. In view of 4.6, it remains to check that one has:

$$(x) \quad A_m \cdot A_{-m} = A_0 \quad \text{for every } m \in M.$$

One also verifies by an immediate calculation that (x) is equivalent to saying that $P \times_S G \rightarrow P \times_S P$ is a *closed immersion*, and not merely a monomorphism (according to the hypothesis that G acts freely), i.e. that the homomorphism on affine rings

$$\theta : A \otimes_{A_0} A \rightarrow A(M)$$

is surjective.⁽¹⁸⁾ This gives 5.1 when one supposes that the equivalence relation defined by G in P is closed. We shall, however, show that this hypothesis is already a consequence of the fact that G acts freely (which is, moreover, implicitly contained in theorem 5.1, since $G \times_S P$ must be isomorphic to $P \times_X P$, which is closed in $P \times_S P$ since X is affine over S and hence separated over S).

Let $R = P \times_S G$. The hypothesis that G acts freely, i.e. that $R \rightarrow P \times_S P$ is a monomorphism, is expressed by saying that the diagonal morphism

$$R \rightarrow R \times_{(P \times_S P)} R = R'$$

is an isomorphism. One has $R = \text{Spec}(A(M))$ and

$$R' = \text{Spec}(A(M \times M)/K)$$

⁽¹⁹⁾ where K is the ideal generated by the elements of the form

$$x_m(e_{m,0} - e_{0,m}), \quad \text{with } m \in M, x_m \in A_m;$$

let $\phi : A(M \times M) \rightarrow A(M)$ be the surjective ring homomorphism defined by

$$xe_{m,n} \mapsto xe_{m+n} \quad (m, n \in M, x \in A)$$

(where the e_m , resp. $e_{m,n} = e_m \otimes e_n$, are the elements of the canonical basis of $A(M)$, resp. $A(M \times M)$). Then the diagonal morphism $R \rightarrow R'$ corresponds to the homomorphism

$$\bar{\phi} : A(M \times M)/K \rightarrow A(M)$$

obtained by passing to the quotient by K . Now the kernel of ϕ is the ideal K' generated by the

$$d_m = e_{m,0} - e_{0,m}.$$

One has $K \subseteq K'$, and the hypothesis that G acts freely on P , i.e. that $\bar{\phi}$ is an isomorphism, is equivalent to the equality $K' = K$, which is expressed by the relations

$$(xx) \quad d_m \in K = \sum_p A(M \times M) A_p d_p, \quad \text{for every } m \in M.$$

Using the natural trigrading of $A(M \times M)$, and the fact that the first degree of d_m is zero, this means that one can write d_m as a sum of elements of the form

$$fe_{r,s}(e_{p,0} - e_{0,p}) \quad \text{with } f \in A_{-p} \cdot A_p,$$

and, using the fact that the total degree of d_m is m , one may restrict to terms such that

$$r + s + p = m.$$

⁽¹⁸⁾**N.D.E.** Indeed, for every $b \in A$, $a_m \in A_m$, one has $\theta(b \otimes a_m) = ba_m \otimes e_m$, so surjectivity of θ is equivalent to (x).

⁽¹⁹⁾**N.D.E.** The ideal denoted by J in the original has been denoted by K , to distinguish it from the ideals J_p of A_0 which appear in (xxx). On the other hand, it has been made explicit below that relations (xx) are equivalent to the equality $K = K'$ (see what follows).

Thus one must have, for every $m \in M$, an expression:

$$(xxx) \quad \begin{cases} d_m = e_{m,0} - e_{0,m} = \sum_{r,s} \lambda_{r,s} (e_{m-s,s} - e_{r,m-r}) \\ \text{with } \lambda_{r,s} \in J_p = A_p \cdot A_{-p} \subseteq A_0, \quad p = m - (r + s). \end{cases}$$

One must deduce relation (x), i.e. the relations

$$(xxxx) \quad J_n = A_0 \quad \text{for every } n \in M.$$

Now for this it suffices to establish the same relation modulo every maximal ideal of A_0 . As the hypotheses made are invariant under such a reduction, one may already suppose that A_0 is a *field*.

Lemma 5.2. *Under the preceding conditions (with A_0 a field), if $M \neq 0$, there exists a $p \in M - \{0\}$ such that $J_p = A_0$.*

Indeed, if this were not so, the sum in the last member of (xxx) would be zero for every $m \in M$, which is absurd.

Corollary 5.3. *Under the preceding conditions, but without any longer supposing that A_0 is a field, there exists a finite number of elements $p_i \in M - \{0\}$ such that $\sum_i J_{p_i} = A_0$.*

Indeed, one applies result 5.2 to the situations deduced from A/A_0 by reduction modulo the maximal ideals of A_0 .⁽²⁰⁾

Corollary 5.4. *Suppose again that A_0 is a field. Then for every subgroup N of M such that $N \neq M$, there exists a $p \in M - N$ such that $J_p = A_0$.*

Indeed, let $M' = M/N$, and consider the ring A' graded of type M' whose underlying ring is A and whose grading is given by

$$A'_{m'} = \coprod_{m \in h^{-1}(m')} A_m,$$

where $h : M \rightarrow M' = M/N$ is the canonical homomorphism. Geometrically, this construction amounts to considering the subgroup $G' = D_S(M')$ of G , and the structure on P of a scheme with group of operators G' induced by the operations of G . It is then evident that G' acts freely on P' , i.e. the pair (M', A') satisfies the hypotheses of 5.3. One therefore obtains

$$1 = \sum_i f_i g_i \quad \text{with } f_i \in A'_{m'_i}, \quad g_i \in A'_{-m'_i} \quad \text{and } m'_i \in M' - \{0\},$$

whence immediately conclusion 5.4 by taking the components of the right-hand side in A_0 , and using the fact that A_0 is a field.

Let us now note that

$$J_p \cdot J_q \subseteq J_{p+q} \quad \text{and } J_p = J_{-p},$$

so, if N denotes the set of $m \in M$ such that $J_p = A_0$, one sees that N is a subgroup of M . Using 5.4 one sees that it equals M . This completes the proof of theorem 5.1. $\square$

As we pointed out in the course of the proof, theorem 5.1 implies:

Corollary 5.5. *Under the conditions of 5.1, the graph morphism*

$$P \times_S G \rightarrow P \times_S P$$

is a closed immersion.

One concludes immediately:

Corollary 5.6. *Let σ be a section of P over S . Then the morphism $g \mapsto \sigma \cdot g$ from G to P defined by σ is a closed immersion.*

⁽²⁰⁾**N.D.E.** It follows from 5.2 that $\sum_{p \neq 0} J_p = A_0$, so 1 is written as a finite sum $\sum_i x_i$, with $x_i \in J_{p_i}$.

Corollary 5.7. *Let G, H be two S -groups, with G diagonalizable, H affine over S , and let $u : G \rightarrow H$ be a homomorphism of S -groups which is a monomorphism. Then u is a closed immersion, $H/G = X$ exists and H is a principal homogeneous bundle over X with group G_X ; finally, X is affine over S .*

Corollary 5.8. *Under the conditions of 5.1, if P is of finite type (resp. of finite presentation) over S , the same is true of $X = P/G$.*

Indeed, it follows from the hypothesis that the fibers of G_X are of finite type, so G_X is of finite presentation over X by 2.1 b); hence P , being a torsor under G_X , is of finite presentation over X .⁽²¹⁾ Since it is also faithfully flat over X , our conclusion then follows from Exp. V, Prop. 9.1.

6. Essentially free morphisms, and representability of certain functors of the form $\prod_{Y/S} Z/Y$

(22)

Definition 6.1. Let $f : X \rightarrow S$ be a morphism of preschemes. One says that f is *essentially free*, or again that X is *essentially free over S* , if one can find a covering of S by affine open subsets S_i , for every i an S_i -prescheme S'_i affine and faithfully flat over S_i , and a covering $(X'_{ij})_j$ of $X'_i = X \times_S S'_i$ by affine open subsets X'_{ij} , such that for every (i, j) the ring of X'_{ij} is a free module over the ring of S'_i .⁽²³⁾

Proposition 6.2. *a) If X is essentially free over S , it is flat over S , the converse being true if S is artinian.*

b) If S is the spectrum of a field, every S -prescheme is essentially free over S .

c) If X is essentially free over S , and if $S' \rightarrow S$ is a base change morphism, $X' = X \times_S S'$ is essentially free over S' . The converse is true if $S' \rightarrow S$ is faithfully flat and quasi-compact.

The proof is immediate, using for the converse in a) the fact that a flat module over an artinian local ring is free.⁽²⁴⁾

Proposition 6.3. *Let H be a diagonalizable S -group prescheme (more generally, one which becomes diagonalizable by a suitable faithfully flat quasi-compact extension of every affine open subset of S , i.e. H is “of multiplicative type”, cf. IX 1.1). Then H is essentially free over S .*

Indeed, if H is diagonalizable, it is affine over S and defined by an algebra which is a free $\mathcal{O}_S$ -module.

The introduction of definition 6.1 is justified by the

⁽²¹⁾**N.D.E.** Cf. EGA II, 2.7.1 (vi).

⁽²²⁾The present no. is independent of the theory of diagonalizable groups; its natural place would be in VI_B.

⁽²³⁾**N.D.E.** One may replace “free” by “projective”, cf. 6.8 below. On the other hand, this notion should be compared with that of a flat and pure S -prescheme, introduced and developed in [RG71]; see in particular loc. cit., Part 1, 3.3.12 and Part 2, 3.1.4.1.

⁽²⁴⁾**N.D.E.** Indeed, let $(A, \mathfrak{m})$ be an artinian local ring, k its residue field, M an arbitrary A -module, $(x_i)_{i \in I}$ elements of M whose images form a basis of $M/\mathfrak{m}M$ over k . Let F be the free A -module with basis $(e_i)_{i \in I}$, and $\phi : F \rightarrow M$ the A -morphism defined by $\phi(e_i) = x_i$. Then $Q = \text{Coker } \phi$ satisfies $Q = \mathfrak{m}Q$, whence, since $\mathfrak{m}$ is nilpotent, $Q = 0$. Suppose in addition that M is flat over A ; then $K = \text{Ker } \phi$ satisfies $K \otimes_A k = 0$, i.e. $K = \mathfrak{m}K$, whence $K = 0$.

Theorem 6.4. *Let S be a prescheme, Z an essentially free S -prescheme, Y a closed subprescheme of Z . Consider the following functor⁽²⁵⁾*

$$F = \prod_{Z/S} Y/Z : (\text{Sch})_{/S}^{\circ} \longrightarrow (\text{Ens}), \quad F(S') = \Gamma(Y_{S'}/Z_{S'}) = \begin{cases} \emptyset & \text{if } Z_{S'} \neq Y_{S'}; \\ \{\text{id}_{Z_{S'}}\} & \text{if } Z_{S'} = Y_{S'}. \end{cases}$$

This functor is representable by a closed subprescheme of S .⁽²⁶⁾

⁽²⁷⁾ Let us first note that F is a sheaf for the (fpqc) topology: since $F(S') = \emptyset$ or $\{\text{pt}\}$ for every S' , this amounts to checking that if (S_i) is an open covering of S (resp. $S' \rightarrow S$ a faithfully flat and quasi-compact morphism), and if each $Y_{S_i} \rightarrow Z_{S_i}$ (resp. if $Y_{S'} \rightarrow Z_{S'}$) is an isomorphism, the same is true of $Y \rightarrow Z$; now this is clear (resp. follows from SGA 1, VIII 5.4 or EGA IV₂, 2.7.1). Moreover, by SGA 1, VIII 2.1 and 5.5, faithfully flat and quasi-compact morphisms are of effective descent for the fibered category of closed immersion arrows. This allows us to restrict, with the notation of 6.1, to the case where $S = S'_i$.

Let then (Z_j) be a covering of Z by affine open subsets such that $\mathcal{O}(Z_j)$ is a free module over $A = \mathcal{O}(S)$, and let $Y_j = Y \cap Z_j$ and $F_j : (\text{Sch})_{/S}^{\circ} \longrightarrow (\text{Ens})$ be the functor defined in terms of (Z_j, Y_j) as F is in terms of (Z, Y) . It is a subfunctor of the final functor, and one evidently has $F = \bigcap_j F_j$, which reduces us to proving that each F_j is representable by a closed subscheme T_j of S (for then F will be representable by the closed subscheme T which is the intersection of the T_j). One may therefore suppose Z also affine, $Z = \text{Spec}(B)$, where B is a free A -module. Let J be a subset of B defining the subscheme Y of Z , and let K be the ideal in A generated by the $u_i(J) \subseteq A$, where the $u_i : B \rightarrow A$ are the coordinate forms with respect to the chosen basis. One verifies immediately that $T = \mathcal{V}(K) = \text{Spec}(A/K)$ satisfies the desired condition, which completes the proof.

Examples 6.5. Let us give important examples of functors that reduce to functors $\prod_{Z/S} Y/Z$ of the type considered in 6.4 and for which it will be useful in what follows to have criteria of representability. Denote by S a prescheme, by X, Y, Z , etc., preschemes over S .

a) Let us give ourselves an S -morphism

$$(x) \quad q : X \longrightarrow \underline{\text{Hom}}_S(Y, Z),$$

("X acts on Y, with values in Z"), i.e. a morphism

$$(xx) \quad r : X \times_S Y \longrightarrow Z.$$

Consider a subprescheme Z' of Z , whence a monomorphism

$$\underline{\text{Hom}}_S(Y, Z') \longrightarrow \underline{\text{Hom}}_S(Y, Z)$$

which makes the first functor a subfunctor of the second; let X' be the inverse image of this subfunctor by (x). It is the subfunctor of X such that $X'(T)$ is the set of $x \in X(T)$ such that $q(x) : Y_T \rightarrow Z_T$ factors through Z'_T . This functor X' may be described as follows: put $P = X \times_S Y$, let P' be the inverse image of Z' by $r : P \rightarrow Z$; then one has an evident isomorphism

$$(xxx) \quad X' \simeq \prod_{P/X} P'/P.$$

One therefore obtains: *if Y is essentially free over S and Z' closed in Z , the subfunctor X' of X is representable by a closed subprescheme of X .*

b) Let us give ourselves two ways of making X act on Y with values in Z , i.e. two morphisms

$$q_1, q_2 : X \rightrightarrows \underline{\text{Hom}}_S(Y, Z),$$

⁽²⁵⁾**N.D.E.** Cf. II.1, where this functor is denoted by $\prod_{Z/S} Y$.

⁽²⁶⁾**N.D.E.** Z has been corrected to S .

⁽²⁷⁾**N.D.E.** The original has been expanded in what follows; see also 1.7.3.

and put $X' = \text{Ker}(q_1, q_2)$: it is the subfunctor of X such that $X'(T)$ is the set of $x \in X(T)$ such that the two morphisms $q_1(x), q_2(x) : Y_T \rightrightarrows Z_T$ are equal. Now giving q_1, q_2 is equivalent to giving a morphism

$$q : X \longrightarrow \underline{\text{Hom}}_S(Y, Z \times_S Z),$$

or again a morphism $r : X \times_S Y \longrightarrow Z \times_S Z$; put then $U = Z \times_S Z$, let U' be the diagonal subprescheme of $Z \times_S Z$; then X' is precisely the inverse image of the subfunctor $\underline{\text{Hom}}_S(Y, U') \longrightarrow \underline{\text{Hom}}_S(Y, U)$ by q , and so can be put in the form (xxx), with $P = X \times_S Y$, and $P' =$ inverse image of the diagonal by r , i.e. kernel of $X \times_S Y \xrightarrow[r_2]{r_1} Z$. One is therefore under the conditions of (a).

One thus sees that: *if Y is essentially free over S and Z separated over S , then the subfunctor X' of X is representable by a closed subprescheme of X .*

c) Let us give ourselves a morphism

$$q : X \longrightarrow \underline{\text{Hom}}_S(Y, Y),$$

i.e. “ X acts on Y ”. Let X' be the “kernel” of this morphism, i.e. the subfunctor X' of X such that $X'(T)$ is the set of $x \in X(T)$ such that $q(x) : Y_T \longrightarrow Y_T$ is the identity. This functor is covered by b), as one sees by introducing a second homomorphism

$$q' : X \longrightarrow \underline{\text{Hom}}_S(Y, Y)$$

“by making X act trivially on Y ”. Thus: *if Y is essentially free over S and separated over S , the kernel subfunctor of q is representable by a closed subprescheme of X .*

d) Under the conditions of c), consider the subfunctor Y' of Y “of invariants under X ”, so $Y'(T)$ is the set of $y \in Y(T)$ such that the corresponding morphism $\bar{q}(y) : X_T \longrightarrow Y_T$ is “the constant T -morphism with value y ”. Introducing q' as in c), and the homomorphisms corresponding to q and q' :

$$\bar{q}, \bar{q}' : Y \rightrightarrows \underline{\text{Hom}}_S(X, Y),$$

one sees that Y' is precisely $\text{Ker}(\bar{q}, \bar{q}')$, and is therefore again covered by b) (with the roles of X, Y reversed and $Z = Y$).

Consequently, *if X is essentially free over S , Y separated over S , then the subfunctor Y' of Y of invariants under X is representable by a closed subprescheme of Y .*

e) Constructions of the type made explicit in the preceding examples are especially frequent in group theory. Thus, when G is an S -group prescheme acting on the S -prescheme X :

$$q : G \longrightarrow \underline{\text{Aut}}_S(X),$$

the kernel of q (“the subgroup of G acting trivially”) is a closed subscheme of G provided that X is essentially free and separated over S (example c)), and the subobject X^G of invariants is a closed subprescheme of X , provided that G is essentially free over S and X separated over $S^{(28)}$ (example d)).

Let Y, Z be subpreschemes of X ; consider the subfunctor $\underline{\text{Transp}}_G(Y, Z)$ of G (“transporter of Y into Z ”) whose points with values in a T over S are the $g \in G(T)$ such that the corresponding automorphism of X_T satisfies $g(Y_T) \subseteq Z_T$, i.e. induces a morphism $Y_T \longrightarrow Z_T$ factoring as $Y_T \longrightarrow Z_T$. Thus: *if Y is essentially free over S and Z closed in X , then $\underline{\text{Transp}}_G(Y, Z)$ is a closed subprescheme of G (example a)).*

One may also consider the *strict transporter of Y into Z* ,⁽²⁹⁾ whose points with values in a T over S are the $g \in G(T)$ such that $g(Y_T) = Z_T$, which is precisely $\underline{\text{Transp}}_G(Y, Z) \cap \sigma(\underline{\text{Transp}}_G(Z, Y))$, where σ is the symmetry of G . Consequently, *if Y and Z are essentially free over S and closed in X , the strict transporter of Y into Z is a closed subprescheme of G .*

An important case is that where $X = G$, G acting on itself by inner automorphisms. If H is a subprescheme of G , the strict transporter of H into H is also called the *normalizer of H*

⁽²⁸⁾**N.D.E.** S_X has been corrected to S .

⁽²⁹⁾**N.D.E.** Denoted by $\underline{\text{Transp}}_G^{\text{str}}(Y, Z)$.

in G , and denoted by $\underline{\text{Norm}}_G H$. Thus: if H is a closed subgroup prescheme of G , essentially free over S , then $\underline{\text{Norm}}_G H$ is representable by a closed subgroup prescheme of G .

Finally, let Z be a subprescheme of G ; then its *centralizer* $\underline{\text{Centr}}_G(Z)$ in G is the group subfunctor of G defined by the procedure of d), when one considers that “ Z acts on G ” by the operations induced by those of G ; thus if Z is essentially free over S and G separated over S , $\underline{\text{Centr}}_G(Z)$ is a closed subgroup prescheme of G . In particular, if G is essentially free and separated over S , then the center C of G , which is precisely $\underline{\text{Centr}}_G(G)$, is a closed subgroup prescheme of G .

When S is the spectrum of a field, 6.3 b) shows that in examples a) to e) above the “essentially free” conditions are automatically satisfied; only separation conditions remain. Recalling that a group prescheme over a field is necessarily separated, one finds, for example:

Corollary 6.7. ⁽³⁰⁾ *Let G be a group prescheme over a field k . Then:*

- *For every subprescheme Z of G , the centralizer of Z in G is a closed subgroup prescheme of G ; this is in particular the case for the center $\underline{\text{Centr}}_G(G)$ of G .*
- *More generally, if $u, v : X \rightarrow G$ are morphisms of preschemes, $\underline{\text{Transp}}_G(u, v)$ is representable by a closed subprescheme of G .*
- *For two subpreschemes Y, Z of G , with Z closed, $\underline{\text{Transp}}_G(Y, Z)$ is a closed subprescheme of G . If Y is also closed, one has the same conclusion for $\underline{\text{Transp}}_G^{\text{str}}(Y, Z)$.*
- *For every subgroup prescheme⁽³¹⁾ H of G , the normalizer $\underline{\text{Norm}}_G(H)$ is a closed subgroup scheme of G .*

Remark 6.8. ⁽³²⁾ Let A be a commutative ring, M an A -module, $M^\vee = \text{Hom}_A(M, A)$; endow the A -module $\text{End}_A(M)$ with the topology of discrete pointwise convergence, i.e. a neighborhood basis of 0 consists of the following A -submodules, where $n \in \mathbf{N}$ and $m_1, \dots, m_n \in M$:

$$K(m_1, \dots, m_n) = \{u \in \text{End}_A(M) \mid u(m_i) = 0 \text{ for } i = 1, \dots, n\}.$$

One says that M is a *quasi-free A -module* if the image of the canonical morphism $\Theta : M \otimes_A M^\vee \rightarrow \text{End}_A(M)$ contains id_M in its closure, i.e. if the following condition is satisfied:

$$(*) \quad \text{for every } m_1, \dots, m_n \in M, \text{ there exist } x_1, \dots, x_r \in M \text{ and } f_1, \dots, f_r \in M^\vee \\ \text{such that } m_i = \Theta \left(\sum_{s=1}^r x_s \otimes f_s \right) (m_i) = \sum_{s=1}^r f_s(m_i) x_s \text{ for } i = 1, \dots, n.$$

(In this case, $\text{Im } \Theta$ is dense in $\text{End}_A(M)$ since for every $u \in \text{End}_A(M)$ one has $u(m_i) = \sum_{s=1}^r f_s(m_i) u(x_s) = \Theta(\sum_{s=1}^r u(x_s) \otimes f_s)(m_i)$.)

Let us first note that this property is stable under base change. Indeed, let $\phi : A \rightarrow A'$ be a ring morphism, $M' = M \otimes_A A'$, and $m'_1, \dots, m'_n \in M'$; then $m'_i = \sum_j m_{ij} \otimes b_{ij}$ ($m_{ij} \in M$, $b_{ij} \in A'$); by hypothesis, there exist $x_1, \dots, x_r \in M$ and $f_1, \dots, f_r \in \text{Hom}_A(M, A)$ such that $m_{ij} = \sum_s x_s f_s(m_{ij})$ for every i, j . Denote by $\phi \circ f_s$ the image of f_s in $\text{Hom}_A(M, A') = \text{Hom}_{A'}(M, A')$; then for every $i = 1, \dots, n$ one has:

$$\left(\sum_s x_s \otimes \phi \circ f_s \right) (m'_i) = \sum_{s,j} x_s \otimes \phi(f_s(m_{ij})) b_{ij} = \sum_j \left(\sum_s x_s f_s(m_{ij}) \right) \otimes b_{ij} = m'_i,$$

which proves that M' is quasi-free over A' . Let us also note that every projective A -module P is quasi-free (there exist A -morphisms $A^{(I)} \xrightarrow{\pi} P \xrightarrow{\tau} A^{(I)}$ such that $\pi \circ \tau = \text{id}_P$; denote by (e_i) the canonical basis of $A^{(I)}$ and by f_i the linear form $e_i^* \circ \tau$ on P ; if $m_1, \dots, m_n \in P$, there exists a finite subset J of I such that $m_k = \sum_{i \in J} f_i(m_k) \pi(e_i)$ for $k = 1, \dots, n$).

⁽³⁰⁾ **N.D.E.** The original numbering has been preserved: there is no 6.6.

⁽³¹⁾ **N.D.E.** Indeed, over a field k , every subgroup prescheme of G is closed, cf. VI_A, 0.6.1.

⁽³²⁾ **N.D.E.** The original has been expanded in what follows; in particular lemma 6.8.1 has been added.

Then theorem 6.4 remains valid on replacing, in the statement of definition 6.1, the word “free” by “projective” or, more generally, by “quasi-free”. Indeed, proceeding as in the proof of 6.4, one is reduced to proving the

Lemma 6.8.1. *Let M be a quasi-free A -module, N a submodule, F the covariant functor (A -algebras) $\longrightarrow$ (Ens) such that $F(B) = \{\text{pt}\}$ if $M_B = (M/N)_B$, and $F(B) = \emptyset$ otherwise. Then there exists an ideal K of A such that $F(B) = \{\text{pt}\}$ if and only if the morphism $A \longrightarrow B$ factors through A/K , i.e. one has an isomorphism functorial in B :*

$$F(B) \simeq \text{Hom}_{A\text{-alg.}}(A/K, B).$$

PROOF. Let (n_j) be a system of generators of N , and let F_j be the subfunctor of the final functor $\underline{e}$ ($\underline{e}(B) = \{\text{pt}\}$ for every B) corresponding to the submodule An_j . One has $F(B) = \{\text{pt}\}$ if and only if the image of each n_j in M_B is zero, so F is the intersection of the functors F_j . This reduces us to the case where N is generated by an element n .

Let $\phi : A \longrightarrow B$ be an A -algebra; if the image $n \otimes 1$ of n in M_B is zero, then for every $f \in M^\vee = \text{Hom}_A(M, A)$ one has $0 = f(n) \otimes 1 = \phi(f(n))$. On the other hand, since M is quasi-free, there exist $x_1, \dots, x_r \in M$ and $f_1, \dots, f_r \in M^\vee$ such that $n = \sum_s f_s(n)x_s$, whence $n \otimes 1 = \sum_s x_s \otimes \phi(f_s(n))$. It follows that $n \otimes 1 = 0$ if and only if ϕ factors through A/K , where K is the ideal generated by the $f_s(n)$ for $s = 1, \dots, r$. This proves the lemma. $\square$

7. Appendix: On monomorphisms of group preschemes

The result proved in the present no. is unnecessary for the rest of the seminar, except X 8.8 and XV and XVI. It relies essentially on the existence theorem for quotient groups of Exposé VI_A.⁽³³⁾

Corollary 5.7 leads one to ask under what conditions one can assert that a monomorphism $u : G \longrightarrow H$ of S -groups is an immersion, or even a closed immersion.

We have seen in VI_B 1.4.2 that this is so if S is the spectrum of a field, provided G is of finite type over k and H locally of finite type over k . One easily concludes that the same result remains valid if one merely supposes S artinian.⁽³⁴⁾

On the other hand, it is easy to give examples of bijective monomorphisms which are not immersions, S being, for example, the affine line over a field or the spectrum of a discrete valuation ring. One will take, for example, $H = (\mathbf{Z}/2\mathbf{Z})_S$, $G = G_1 \times_S G_2$, where G_1 is the open subgroup of H complementary to the closed point x distinct from 0 of the fiber $H_s \simeq \mathbf{Z}/2\mathbf{Z}$ (where s denotes a fixed closed point of S), and G_2 the closed subscheme of H which is the sum of the subscheme reduced to the unit section and of the reduced closed subscheme defined by the closed point x . One easily verifies that G_2 is indeed stable under multiplication in H , and hence is a group scheme. The immersions $G_i \longrightarrow H$ ($i = 1, 2$) then define a homomorphism of S -groups $G = G_1 \times G_2 \longrightarrow H$, which is evidently a bijective monomorphism (and in addition a local immersion), but is not an immersion. (One verifies that G and H are reduced, G having three disjoint irreducible components, whereas H has only two irreducible components.) One will note that $G_1 \longrightarrow H$ also gives an example of an open subgroup G of H which is not closed (contrary to what occurs for algebraic groups over a field). The theory of degeneration of elliptic curves supplies other examples of this latter phenomenon, with H in addition smooth over S of relative dimension 1, and G with connected fibers. It is possible,⁽³⁵⁾ on the other hand, that as soon as one supposes G flat over S , and (say) G and H of finite presentation over S , a monomorphism $u : G \longrightarrow H$ of

⁽³³⁾N.D.E. Cf. Exp. VI_A, Theorems 3.2 and 3.3.2.

⁽³⁴⁾N.D.E. Take account of the additions made in VI_B...

⁽³⁵⁾In fact, M. Raynaud has constructed a counterexample, with G smooth with connected fibers, cf. XVI 1.1 c). If one does not suppose G to have connected fibers, one may take $S = \text{Spec}(\mathbf{Z}_2)$, $G = (\mathbf{Z}/2\mathbf{Z})_S$ with the non-neutral closed point removed, $H = (\mu_2)_S$.

S -groups is automatically an immersion. We shall prove a result of this kind, under additional hypotheses.

Let us first note that one may suppose S affine, and (thanks to the finite presentation hypothesis on G and H , which allows one to reduce to the case of the spectrum of a ring of finite type over $\mathbf{Z}$ ⁽³⁶⁾) S noetherian. Then G and H are noetherian. Saying that $u : G \rightarrow H$ is a closed immersion (resp. an immersion) then amounts to saying that u is a monomorphism (which is true by hypothesis) and that u is proper (resp. and that u is proper at every point of $u(G)$).⁽³⁷⁾ The valuative criterion of properness assures us that it suffices to check that for every base change $S' \rightarrow S$, with S' the spectrum of a discrete valuation ring, complete if one wishes, the morphism $u' : G' \rightarrow H'$ has the same properness property. (The case of local properness was forgotten in EGA II 7.3, and will appear as an afterthought in EGA IV.⁽³⁸⁾) This therefore reduces us to the case where S is itself the spectrum of a complete discrete valuation ring, — provided that the additional hypotheses on G, H, u which we shall be led to formulate are stable under base change.

Let then s (resp. s') be the closed point (resp. the generic point) of S . Then the homomorphisms induced on the fibers

$$u_s : G_s \rightarrow H_s \quad \text{and} \quad u_{s'} : G_{s'} \rightarrow H_{s'}$$

are closed immersions, since they are monomorphisms of group schemes of finite type over fields (VI_B 1.4.2). We may therefore identify $G_{s'}$ with a closed subscheme of $H_{s'}$. Now one has the following result:

Lemma 7.1. *Let S be the spectrum of a discrete valuation ring, s' its generic point, H an S -prescheme, L' a closed subprescheme of the generic fiber $H_{s'}$, so that L' is also a subprescheme of H .*

Then the schematic closure $\overline{L'}$ in H (i.e. the smallest closed subprescheme of H containing L' , cf. EGA I 9.5) exists and is also the unique closed subprescheme of H , flat over S , whose generic fiber is L' . Moreover, formation of L is functorial with respect to a variable pair (H, L') and commutes with formation of cartesian products over S .

In particular, if H is an S -group and L' a subgroup of $H_{s'}$, then L is a subgroup of H .

The proof is immediate and left to the reader.⁽³⁹⁾ Applying this to the situation $H, G_{s'}$, one sees that the monomorphism $u : G \rightarrow H$ factors as $G \rightarrow L \rightarrow H$, where L is a subgroup of H which is a closed subprescheme, flat over S , and where $G \rightarrow L$ induces an isomorphism on the generic fibers. Then $u : G \rightarrow H$ is an immersion (resp. a closed immersion) if and only if $u' : G \rightarrow L$ is so. This therefore reduces us to the case where H is flat over S and $u_{s'} : G_{s'} \rightarrow H_{s'}$ an isomorphism (provided that the additional hypotheses which we shall have to formulate on G, H, u are respected when one replaces H by a closed subgroup prescheme). As H is then the schematic closure of $H_{s'}$, if u is an immersion, $u(G)$ will be a subprescheme⁽⁴⁰⁾ of H containing $H_{s'}$, so its schematic closure will also be H , and consequently $u(G)$ will be an open subprescheme of H . Consequently, we shall in fact have to prove that u is an open immersion (resp. an isomorphism if we wished to establish that u is a closed immersion). As G and H are flat over S , it amounts to the same thing to say that the morphisms induced on the fibers are open immersions, resp. isomorphisms (cf. SGA 1, I 5.7), and since this is already so for $u_{s'}$, one is reduced to proving that u_s is an open immersion, resp. an isomorphism. Let us first note that, since G and H are flat over S , the dimension of their fibers remains constant (VI_B 4.3). As G and H have the same generic fiber, it follows that this dimension is the same for G and H , so $u_s : G_s \rightarrow H_s$ is a monomorphism

⁽³⁶⁾**N.D.E.** Cf. EGA IV₃, §8, and Exp. VI_B, §10.

⁽³⁷⁾**N.D.E.** Cf. EGA IV₃, 8.11.5 and 15.7.1.

⁽³⁸⁾Cf. EGA IV₃, 15.7.

⁽³⁹⁾Cf. EGA IV₂, 2.8.

⁽⁴⁰⁾**N.D.E.** The word “closed” has been deleted.

of algebraic groups of the same dimension. One easily concludes that G_s is open in H_s , and in fact is set-theoretically a union of connected components of H_s (one is reduced to the case where the base field is algebraically closed and G and H reduced, hence smooth over k , where this is immediate. . .). Thus $H_s - G_s$ is closed in H_s , hence in H , so its complement H' in H is open, and is evidently an open subset stable under the group law of H . Thus, replacing H by H' if necessary, to prove that u is an immersion one may reduce (with the usual proviso) to the case where, in addition to the preceding hypotheses, one supposes u bijective, i.e. $u_s : G_s \rightarrow H_s$ bijective. One is therefore in any case reduced to proving that u , or again u_s , is an isomorphism, with the bijectivity hypothesis when appropriate.

Let us therefore first suppose that u is bijective. If H_s is reduced, one may evidently conclude that u_s is an isomorphism, for G_s is identified with a closed subscheme of H_s with the same underlying set. In particular, if $k = \kappa(s)$ is of characteristic zero, every algebraic group over k is reduced by Cartier (cf. VI_B, 1.6.1, or VII_B, 3.3.1, or EGA IV₄, 16.12.2 and 17.12.5), and one has thus obtained:

Proposition 7.2. *Let $u : G \rightarrow H$ be a homomorphism of group preschemes of finite presentation over S . Suppose that u is a monomorphism, G flat over S , and the residue fields of S of characteristic zero. Then u is an immersion.*

When $\kappa(s)$ is of characteristic $p > 0$, we shall restrict to the case where G is commutative. Then (with the reductions made) H is also commutative, since it is the schematic closure of $H_{s'}$, which is isomorphic to $G_{s'}$ and hence commutative. For every integer $n \geq 0$, we put $S_n = \text{Spec}(V/\mathfrak{m}^{n+1})$ (where V is the valuation ring defining S and $\mathfrak{m}$ its maximal ideal), $G_n = G \times_S S_n$, $H_n = H \times_S S_n$. For every integer $m > 0$, we also introduce the subgroups ${}_m G$ and ${}_m H$ of G and H , kernels of the m -th power. One defines similarly ${}_m(G_n) = ({}_m G)_n$, which one will denote simply by ${}_m G_n$, and similarly for H .

By VI_A 3.2, one may form the quotients $Q_n = H_n/G_n$; then Q_n is a commutative group scheme over S_n , flat over S_n , and $H_n \rightarrow Q_n$ is a faithfully flat morphism with kernel G_n . As formation of quotients commutes with extension of the base,⁽⁴¹⁾ one has

$$Q_n \simeq Q_m \times_{S_m} S_n \quad \text{for } m \geq n,$$

in particular the fiber $Q_n \times_{S_n} S_0$ is precisely $Q_0 = H_0/G_0$. As G_0 has the same underlying set as H_0 , Q_0 is set-theoretically reduced to a single point: it is a purely infinitesimal group. Consequently, each Q_n is finite and flat over S_n . Thus Q_n is defined by an algebra C_n over $V_n = V/\mathfrak{m}^{n+1}$ which is a free module of finite type over this ring, and, for $m \geq n$, one has $C_n = C_m \otimes_{V_m} V_n$, an isomorphism also respecting the diagonal map. One therefore obtains a free module of finite type $C = \varprojlim C_n$ over $V = \varprojlim V_n$, and the diagonal maps of the C_n define a diagonal map of C , so that $Q = \text{Spec}(C)$ becomes a finite and flat group scheme over S such that

$$Q \times_S S_n = Q_n$$

for every n .

Lemma 7.3. ⁽⁴²⁾ *Let K be a field, Q a finite group scheme over K , of degree n . Then the n -th power morphism in Q is zero.*

Cf. VII_A 8.5.

Remark 7.3.1. Statement 7.3 still makes sense for a group scheme Q/S finite and locally free over S , S being any base prescheme. It would be interesting to find a proof in this general case.

⁽⁴¹⁾**N.D.E.** Bring this out in Exp. V and VI_A...

⁽⁴²⁾**N.D.E.** The hypothesis that Q is commutative has been deleted here, and 7.3.1 has been modified accordingly. Note that if Q is not commutative, the n -th power morphism is not in general a group homomorphism.

One will note that 7.3 (i.e. VII_A, 8.5) proves in any case that the contemplated statement is true if S is reduced, as one sees by applying 7.3 to the fibers of Q at the maximal points (i.e. generic points of the irreducible components) of S .⁽⁴³⁾

In particular, under the conditions of the preceding proof, where S is the spectrum of a discrete valuation ring and Q is commutative, one finds that $n \cdot \text{id}_Q = 0$. Moreover, here n is a power p^{ν_0} of the residue characteristic, and one finds the

Corollary 7.4. *Q and the Q_n being as above, and their common degree being p^{ν_0} , one shall have $p^{\nu_0} \cdot \text{id}_Q = 0$ and consequently $p^{\nu_0} \cdot \text{id}_{Q_n} = 0$ for every n .*

Corollary 7.5. *Suppose in addition G_0 smooth over k , and the homomorphism $p \cdot \text{id}_{G_0}$ flat (which amounts to saying, by the structure of algebraic groups over an algebraically closed field $\bar{k}$, that $G_{\bar{k}}$ contains no subgroup isomorphic to the additive group). Then for every $\nu \geq \nu_0$ and $n > 0$, the group $p^\nu H_n$ is flat over S_n .*

Indeed, it follows from $p^{\nu_0} \cdot \text{id}_{Q_n} = 0$ that $p^\nu \cdot \text{id}_{H_n}$ factors through G_n , so one shall have a commutative diagram:

$$\begin{array}{ccc} H_n & \xrightarrow{p^\nu \cdot \text{id}_{H_n}} & H_n \\ u_n \uparrow & \searrow v_n & \uparrow u_n \\ G_n & \xrightarrow{p^\nu \cdot \text{id}_{G_n}} & G_n \end{array}$$

I say that v_n is flat. Indeed, since H_n and G_n are flat over S_n , one is reduced to checking that v_0 is flat (SGA 1, IV 5.9), so one may suppose $n = 0$, whence $S_n = \text{Spec}(k)$. As $p \cdot \text{id}_{G_0}$, and hence $p^\nu \cdot \text{id}_{G_0}$, is flat, its image is an induced open subgroup G'_0 of G_0 , and as $u_0 : G_0 \rightarrow H_0$ is surjective, it follows that v_0 takes its values in G'_0 , and hence may be regarded as a homomorphism into G'_0 . As its composite with u_0 is an epimorphism, it is an epimorphism, hence a flat homomorphism into G'_0 , and hence a flat homomorphism into G_0 . Thus v_n is flat, so $\text{Ker } v_n = p^\nu H$ is flat over S . Q.E.D.

Remark. We have not explicitly used the fact that G_0 is smooth over k ; but it is easy to see that this is a consequence of the fact that $p \cdot \text{id}_G$ is flat, which is why we made this condition explicit in the hypothesis of corollary 7.5.

Lemma 7.6. *Let $u : G \rightarrow H$ be a surjective monomorphism of commutative algebraic groups over a field k of characteristic $p > 0$; consider the (purely infinitesimal) group $Q = H/G$. Then there exists an integer ν_1 such that for $\nu \geq \nu_1$ the sequence*

$$0 \rightarrow p^\nu G \rightarrow p^\nu H \rightarrow Q \rightarrow 0$$

is exact.

It suffices to ensure exactness at Q , and for this to ensure that the homomorphism

$$(x) \quad \mathcal{O}_{Q,e} \rightarrow \mathcal{O}_{p^\nu H,e}$$

of local rings at the neutral elements is injective (N.B. recall that Q is set-theoretically reduced to the element e). Now one has a natural homomorphism

$$(xx) \quad \mathcal{O}_{p^\nu H,e} \rightarrow \mathcal{O}_{H,e}/\mathfrak{m}^{p^\nu},$$

where $\mathfrak{m}$ is the augmentation ideal (i.e. the maximal ideal) of $\mathcal{O}_{H,e}$, as one sees by noting that $p^\nu \cdot \text{id}_H$ vanishes on the kernel of the “iterated Frobenius homomorphism” F^ν ; the composite of homomorphisms (x) and (xx) also equals the natural composite

$$(xxx) \quad \mathcal{O}_{Q,e} \rightarrow \mathcal{O}_{H,e} \rightarrow \mathcal{O}_{H,e}/\mathfrak{m}^{p^\nu}.$$

⁽⁴³⁾**N.D.E.** Add this in VII_A — On the other hand, the statement is true for every S if Q is commutative, cf. Deligne’s theorem in [TO70], p. 4.

Now $\mathcal{O}_{Q,e} \rightarrow \mathcal{O}_{H,e}$ is injective, since $H \rightarrow Q$ is an epimorphism and hence flat; on the other hand $\mathcal{O}_{Q,e}$ is artinian; finally, the intersection in $\mathcal{O}_{H,e}$ of the $\mathfrak{m}^{p^\nu}$ is reduced to 0, and hence so is the intersection of their traces on $\mathcal{O}_{Q,e}$. Consequently, one of these traces is reduced to 0, which proves that (xxx) is injective, and a fortiori (x) is injective.

Lemma 7.7. *Under the conditions of 7.5 there exists a ν_1 such that, for $\nu \geq \nu_1$ and every n , the sequence of S_n -groups*

$$0 \rightarrow {}_{p^\nu}G_n \rightarrow {}_{p^\nu}H_n \xrightarrow{w_n} Q_n \rightarrow 0$$

is exact (more precisely, w_n is faithfully flat and its kernel is ${}_{p^\nu}G_n$).

One takes ν_1 as in 7.6 applied to $u_0 : G_0 \rightarrow H_0$, and $\nu_1 \geq \nu_0$ (where $p^{\nu_0} = \text{rank } Q_0$). One need only check that w_n is faithfully flat. Now by 7.5, ${}_{p^\nu}H$ is flat over S_n , and since Q_n is also so, one is reduced to checking that $w_0 : {}_{p^\nu}H_0 \rightarrow Q_0$ is faithfully flat, i.e. is an epimorphism, which is true by the choice of ν_1 .

Corollary 7.8. *Suppose in addition H separated over S , more generally that ${}_{p^\nu}H$ is separated over S for every ν , and that ${}_{p^\nu}G$ is finite over S for every ν . Then the group schemes ${}_{p^\nu}G$ and ${}_{p^\nu}H$ are finite and flat over S for $\nu \geq \nu_1$, and one has an exact sequence*

$$0 \rightarrow {}_{p^\nu}G \rightarrow {}_{p^\nu}H \xrightarrow{w} Q \rightarrow 0.$$

As $u : G \rightarrow H$ is surjective, so is the induced morphism ${}_{p^\nu}G \rightarrow {}_{p^\nu}H$, and as the first member is finite over S and the second separated over S , it follows that the second member is finite over S . To check then that ${}_{p^\nu}G$ and ${}_{p^\nu}H$ are also flat over S , it suffices to check that this is so for ${}_{p^\nu}G_n$ and ${}_{p^\nu}H_n$ for every n , which is contained in 7.5. Finally, the exact sequence of 7.8 comes from the exact sequences 7.7 for variable n .

Taking the generic fibers in exact sequence 7.8 and recalling that $u_{s'} : G_{s'} \rightarrow H_{s'}$ is an isomorphism, one finds $Q_{s'} =$ the unit group, whence (since Q is flat over S) $Q =$ the unit group, whence $Q_s =$ the unit group, so $u_s : G_s \rightarrow H_s$ is an isomorphism. Thus:

Proposition 7.9. *Let $u : G \rightarrow H$ be a homomorphism of group preschemes of finite presentation over the prescheme S . Suppose:*

- a) *u is a monomorphism.*
- b) *G is flat over S .*
- c) *For every $s \in S$ such that $\kappa(s)$ is of residue characteristic $p > 0$, one requires that the following conditions be satisfied for the homomorphism $u' : G' \rightarrow H'$ of group preschemes over $S' = \text{Spec}(\mathcal{O}_{S,s})$ deduced from $u : G \rightarrow H$ by the base change $S' \rightarrow S$:*

- *G' is commutative,*
- *the special fiber $G'_0 = G'_s$ is smooth over $\kappa(s)$,*
- *for every integer $\nu > 0$, ${}_{p^\nu}G'$ is finite over S' and ${}_{p^\nu}H'$ is separated over S' .*

Under these conditions, u is an immersion.

It suffices to note that in c), the condition that ${}_{p^\nu}G'$ be finite over S' implies that ${}_{p^\nu}G'_0$ is finite over $\kappa(s)$, which already implies that $G'_0 \otimes_{\kappa(s)} \overline{\kappa(s)}$ has no subgroup isomorphic to the additive group, so one is under the hypotheses of 7.5.

Remark 7.10. Examples of M. Raynaud (XVI 1.1 a) and b)) show that in c) one can drop neither the hypothesis that the ${}_{p^\nu}G'$ are finite over S' nor that the ${}_{p^\nu}H'$ are separated over S' .

We now want conditions ensuring that u is a closed immersion. We therefore retain the hypotheses preceding 7.9, but no longer suppose u surjective (only that u is an isomorphism on the generic fibers). We already know that $u : G \rightarrow H$ is an open immersion; the same is therefore true of $u_0 : G_0 \rightarrow H_0$, whose image thus contains the connected component

of the neutral element $(H_0)^0$. Note that since G_0 is smooth over k and “without additive component” over the algebraic closure of k , the same is true of H_0 . Now one has the

Lemma 7.11. *Let H be a commutative algebraic group over a field k , such that $H \otimes_k \bar{k}$ contains no subgroup isomorphic to $\mathbf{G}_a$. Let $n = \text{degree of } H/H^0$; then the homomorphism*

$${}_nH \longrightarrow H/H^0$$

is surjective.

Indeed, one may suppose k algebraically closed, and then this follows from the well-known fact that $H^0(k)$ is a divisible group.

Suppose now (returning to our situation $u : G \longrightarrow H$) that the ${}_nG$ ($n > 0$) are proper over S , and the ${}_nH$ are separated over S . Let us note, on the other hand, that the ${}_nH$ are flat over S . Indeed, it suffices to see this at the points over s ; one is then reduced to proving that $n \cdot \text{id}_H$ is flat at the points over s , and for this one is reduced to checking that $n \cdot \text{id}_{H_0}$ is flat, which is equivalent (as we have already noted) to the fact that $(H_0)^0$ is smooth⁽⁴⁴⁾ over k and has no $\mathbf{G}_a$ component over the algebraic closure of k . As $(H_0)^0 = (G_0)^0$, this follows from the analogous hypothesis made on G_0 . On the other hand the ${}_nH$ are separated over S since H is so, hence the morphism ${}_nG \longrightarrow {}_nH$ is proper, hence its image is closed. As this image contains the generic fiber of ${}_nH$ (since $u_{s'} : G_{s'} \longrightarrow H_{s'}$ is an isomorphism) and ${}_nH$ is flat over S , hence identical to the closure of its generic fiber, it follows that ${}_nG \longrightarrow {}_nH$ is surjective, hence ${}_nG_0 \longrightarrow {}_nH_0$ is surjective. Recalling that $G_0 \supseteq (H_0)^0$ and applying 7.11, one finds that $G_0 \longrightarrow H_0$ is surjective, hence u is surjective. One thus obtains:

Proposition 7.12. *With the notation of 7.9, suppose that conditions a) and b) are satisfied, as well as condition c') (stronger than c)), obtained by requiring that for every integer $n > 0$ (not only of the form p^ν), ${}_nG'$ be finite over S' and ${}_nH'$ separated over S' . Under these conditions, u is a closed immersion.*

Remark 7.13. a) One easily checks that the separation hypothesis made on the ${}_nH$ in fact implies that H is separated over S . This is moreover formally contained in 7.12 by taking for G the unit S -group. Let us also note that when S is locally noetherian, one may in 7.9 restrict to supposing H locally of finite type over S (instead of: of finite type over S), the proof given applying as it stands; for 7.12 one shall suppose in addition that the fibers of H are of finite type.

b) Using 7.12, it is not difficult to prove that if $u : G \longrightarrow H$ is a monomorphism of S -group preschemes of finite presentation, with G diagonalizable (or more generally “of multiplicative type”) and H separated over S , then u is a closed immersion, — which constitutes a satisfactory generalization of the first conclusion stated in 5.7. When G is smooth over S , it suffices to apply 7.12, and the general case reduces easily to that one. When one no longer supposes H separated over S , one can still show that u is an immersion; in the case where G is a torus, this fact follows, moreover, also from what follows.

c) When in 7.9 one supposes G to have connected fibers, one may in condition 7.9 c) drop the hypothesis that the ${}_p H'$ are separated over S' . Indeed, with the reductions made in the proof, one may suppose H flat over S and u bijective, hence H with connected fibers, hence H separated over S by Raynaud’s theorem (VI_B 5.5).⁽⁴⁵⁾

⁽⁴⁴⁾**N.D.E.** The terminology has been updated, replacing “simple” by “smooth” (see, for example, note (1) of A. Grothendieck in SGA 1, Exp. II).

⁽⁴⁵⁾**N.D.E.** Specify this attribution in VI_B §5, which appeared in SGAD 1965, indicating the modifications between SGAD and Lect. Notes 151.

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⁽⁴⁶⁾**N.D.E.** Additional references cited in this Exposé.